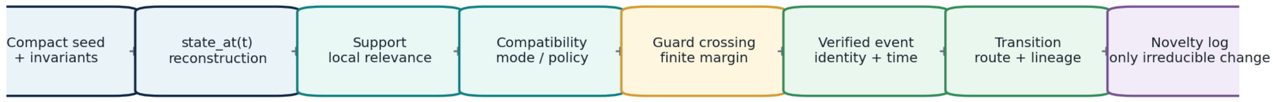
UNIFIED GEOMETRIC-TOPOLOGICAL SUBSTRATE

Societal Applications and Verified GPU Seed-Query Deployment

A practical overview of UGTS-GN 1.1, the full-SGP4 RTX 5070 Ti demonstration, literal use cases, metrics, governance and deployment pathways

UGTS query-first pipeline



Projection, display, actuation and detailed materialization are downstream options - not the state authority.

Canonical shorthand: local support -> compatibility -> guard crossing -> verified event -> route/transition -> lineage + novelty log.

Evidence status

One application has been physically verified: compact orbital seeds reconstructed with full SGP4/SDP4 on an RTX 5070 Ti Laptop GPU, followed by ground-station guard crossing and compact AOS/LOS events. Other societal applications in this report are explicitly labelled as direct extensions, pilot candidates or research candidates.

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Public-facing edition: sensitive personal identifiers intentionally omitted.

Executive summary

Bottom line

The substrate is best understood as a query-first state-and-event architecture. It distributes compact identity-bearing seeds, reconstructs only the requested state, rejects irrelevant or incompatible candidates, detects a finite-margin guard crossing, records the transition and preserves lineage plus irreducible novelty. Its societal value is strongest where dynamics are reconstructible, events are sparse, update bandwidth is limited and decisions can be expressed as explicit guarded transitions.

19.35 M verified seven-day object-time candidates	**91.974 ms** direct full-SGP4 query p50	**717 / 717** exact pass-event identities	**0** propagation and sanitizer errors
5.8 KiB portable KSGP1 seed container	**590.6 MiB** avoided seven-day double4 positions	**210.4 M/s** verified direct screening throughput	**70 C** maximum sustained-run temperature

What has been achieved. The supplied deployment shows that a consumer laptop GPU can reconstruct full near-Earth and deep-space SGP4 state directly from compact seeds and emit identity-preserving acquisition/loss events. It validates the core seed -> state_at(t) -> guard -> event pattern on physical hardware. [V0.5]

What has not been achieved. The result is not a live satellite tracking service, navigation system, collision-avoidance authority, medical alert platform, or proof that every proposed UGTS mechanism works in every domain. The current orbit path still needs fresh authoritative elements, rigorous Earth-orientation handling, uncertainty/covariance and mission certification for operational claims. [V0.5; R5]

Strongest societal opportunities. Near-term opportunities are ground-station and constellation screening, multi-hazard threshold systems, industrial predictive maintenance, grid contingency screening, infrastructure condition events and low-connectivity public-health reporting. The architecture can reduce data movement and operator burden, but those benefits must be measured against false/missed events, model age, energy and equity.

Decision rule. Use direct seed queries when a horizon is queried once, criteria change often, or dense state should not persist. Materialize and cache when the same horizon is queried repeatedly and memory is available. In the verified case, the crossover is just over one repeated query.

Primary architecture basis: UGTS-GN 1.1, especially pp. 2-5 (query-first architecture and identity), p. 10 (precision rule), and p. 15 (kill criteria). Primary measured basis: KLB SeedChain GPU v0.5.0 Laptop Verification, pp. 2-13.

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1. Evidence model and reading rule

The report separates four evidence classes so that an interesting architecture is not mistaken for an operationally certified service.

Class	Meaning	How it is used here
Source-derived	Operators and boundaries stated by UGTS-GN 1.1.	Support, compatibility, guard, event, transition, lineage, novelty, query-first authority and kill criteria.
Measured	Results produced by the supplied RTX 5070 Ti deployment.	SGP4 state accuracy, event identity, latency, throughput, memory avoidance, GPU profile and thermal behavior.
Externally contextualized	Official programmes and standards showing real societal needs.	Early warning, public health surveillance, grid modernization, transport information, space safety and digital twins.
Engineering inference	A domain mapping that has not yet been built or validated.	Use-case designs, pilot metrics, readiness and caveats. These are proposals, not performance claims.

Reading rule

The substrate is not primarily a renderer. Its authoritative product is a query result or verified event. Maps, dashboards, alerts, control actions and visualizations are downstream consumers. [UGTS pp. 2-4]

The public application labels used throughout the report are:

- **VERIFIED APPLICATION:** the supplied code and hardware results directly demonstrate the application.
- **DIRECT EXTENSION:** the same tested kernel pattern can be reused with modest additional engineering.
- **PILOT CANDIDATE:** the mapping is plausible but requires a domain model, independent reference and field test.
- **RESEARCH CANDIDATE:** substantial new modeling, standards, safety or hardware work is required.

Non-negotiable precision boundary

Packed or reconstructed state is acceptable only when coordinate, axis, radius and guard error remain below the declared event margin. A storage reduction is never, by itself, a correctness proof. [UGTS p. 10]

2. The substrate in plain language and formal terms

2.1 ELI5: recipe, bouncer, tripwire and logbook

Analogy	Technical meaning
Recipe	Store a compact recipe for how state evolves instead of saving every possible frame or sample.
Bouncer	Quickly reject candidates outside the local support or with the wrong mode, policy, sheet, identity or capability.
Tripwire	Define a clear guard surface and detect the first valid crossing with a finite error margin.
Logbook	Record what changed, when, why, which route was taken and which identity continued. Store unpredictable novelty; recompute closed dynamics.

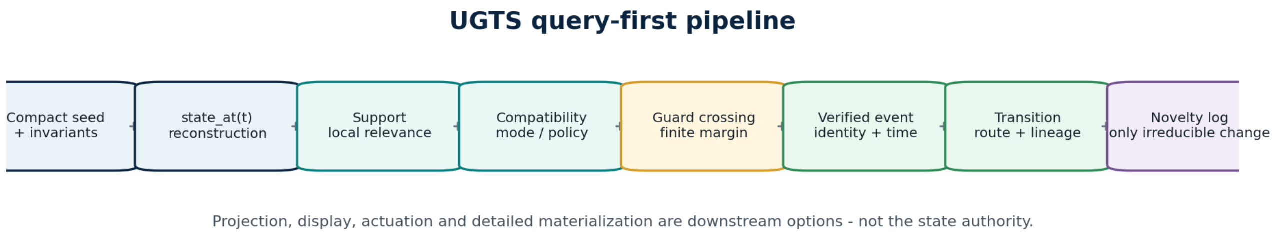

Figure 1. Query-first architecture derived from the canonical UGTS shorthand.

2.2 Formal state-and-event model

State: q = (position, time, phase, sheet, orientation, generative_address, branch, policy, uncertainty).

Query: state_at(t), next_event, events_in_support, or reachable relation under a versioned model.

Verified event: a candidate that passes support and compatibility, crosses a finite-margin guard, satisfies confidence/certification and produces an ordered transition with lineage.

Persistence: identity is not the current coordinate. Closed dynamics may be recomputed from a seed and grammar; exogenous changes must be stored in an ordered novelty log. [UGTS pp. 4-5]

Stage	Question	Failure if omitted
Seed / invariants	What compact state is sufficient to reconstruct the requested model?	Missing state or hidden assumptions make replay impossible.
Support	Which candidates can possibly matter locally?	Expensive work is wasted and the claimed query advantage disappears.
Compatibility	Which co-located or intersecting states are allowed to interact?	Policy, mode, phase or identity errors create false events.
Guard	What scalar boundary and finite margin define the transition?	Noise and quantization change ordering or produce chatter.
Event	Which crossing is valid, earliest and sufficiently certain?	Dense samples replace actionable results.
Transition	What atomic state patch occurs?	Route, branch or mode becomes ambiguous.
Lineage / novelty	Which identity persists and which change could not be reconstructed?	Audits, replay and accountability fail.

3. The verified full-SGP4 application

3.1 Literal application

The verified program is a local satellite visibility and pass-boundary query engine. It consumes a small KSGP1 container holding 32 GPS operational mean-element seeds, initializes the full SGP4/SDP4 branch state, propagates each object through a seven-day horizon at one-second intervals, converts the result to the current simplified Earth-fixed query frame and emits acquisition-of-signal (AOS) and loss-of-signal (LOS) events for a station at 52 N, 5 E.

UGTS role	Verified orbit implementation
Seed / identity	OMM/GP mean elements plus object identity in KSGP1.
state_at(t)	Full near-Earth or deep-space SGP4/SDP4 TEME position and velocity.
Support	Object-time interval inside the declared data horizon.
Compatibility	Object/station query policy; the supplied GPS workload did not add further pruning.
Guard	Ground-station elevation relative to a 10-degree threshold.
Verified event	AOS or LOS boundary with object identity and crossing time.
Lineage	Stable NORAD/object identity and deterministic event lineage.
Novelty	Future OMM/GP updates, station outages or other external

UGTS role	Verified orbit implementation
	changes.

Why no live satellite access was required

The validation question was whether the implementation computes the selected SGP4 model correctly and consistently. Published Vallado vectors, an independent Python SGP4 implementation, CPU/GPU comparison and event reproduction answer that software question. Real-world orbit truth and radio-link performance require fresh tracking data, precise ephemerides or observations and a validated frame/RF stack.

19,353,600 candidate intervals	**19,207,536** support survivors	**5,498,429** visible endpoints	**717** compacted AOS/LOS events
91.974 ms direct query p50	**210.4 M/s** direct throughput	**717 / 717** event identities matched	**0** propagation errors

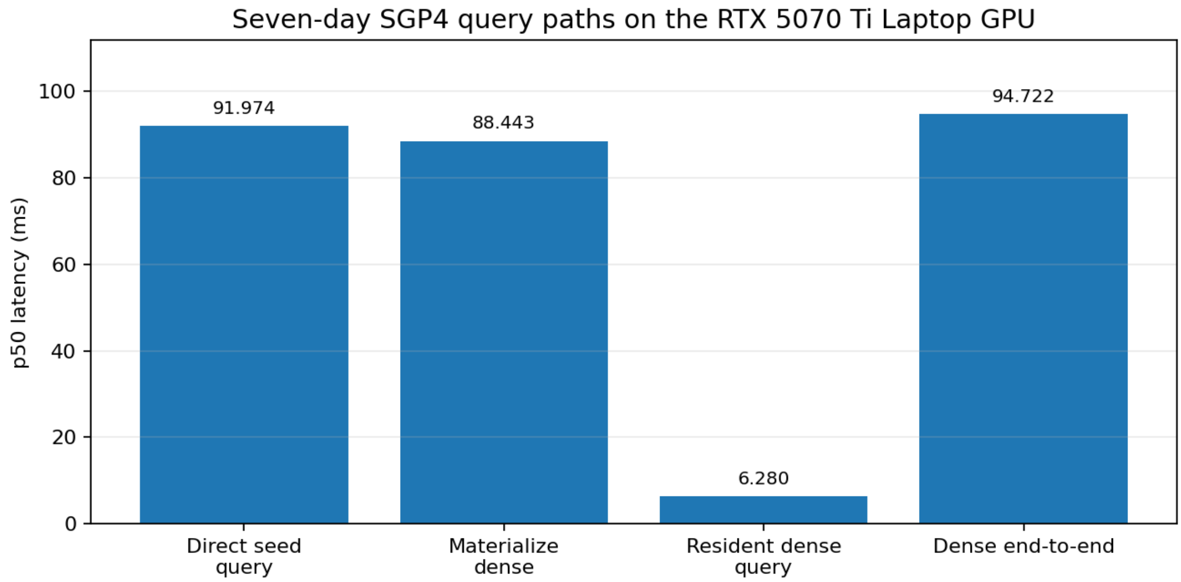

Figure 2. One-pass direct reconstruction narrowly beats materialize-plus-query; a resident dense cache is much faster for repeated queries.

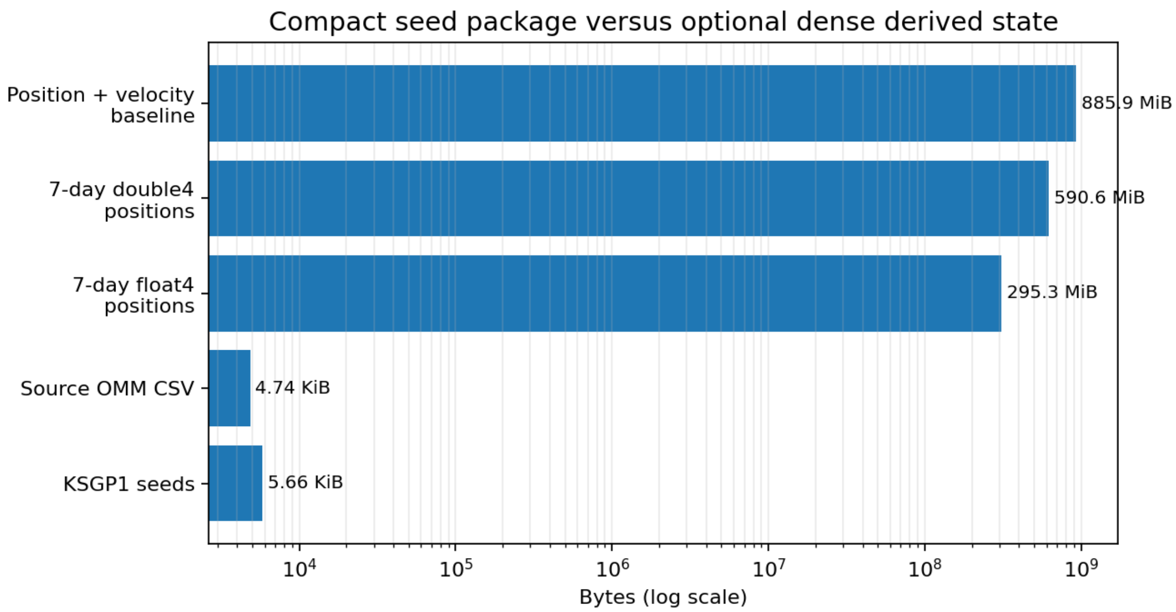

Figure 3. The reported ratio is avoided materialization of derived positions, not ordinary compression of an existing trajectory file.

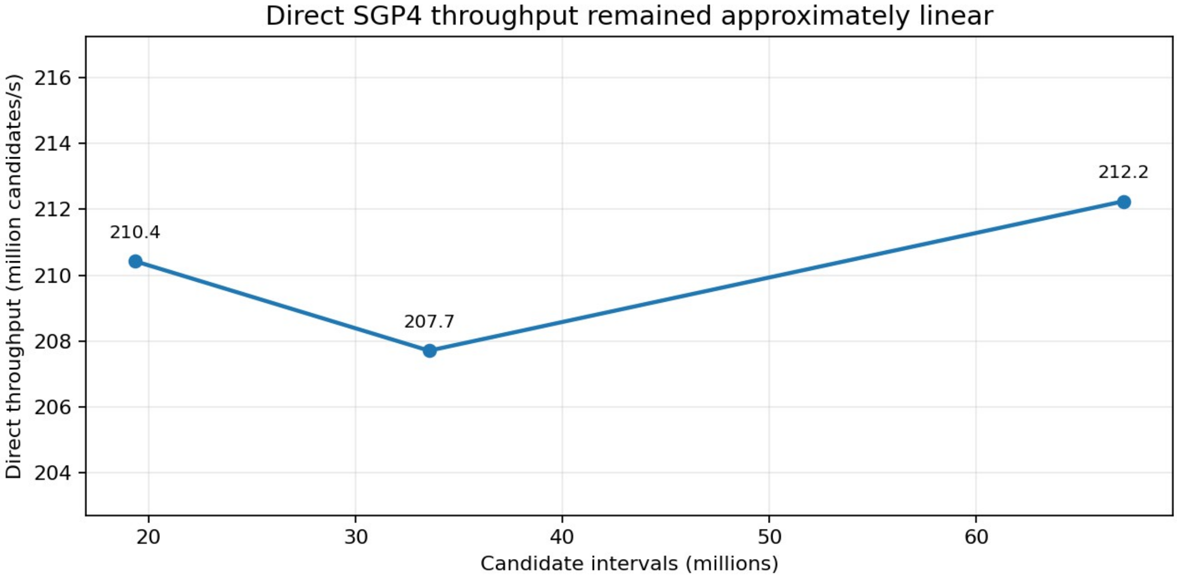

Figure 4. Throughput across the seven-day, 1 GiB and 2 GiB stress workloads.

3.2 What the result proves

- Full SGP4/SDP4 near-Earth, deep non-resonant, synchronous-resonance and half-day-resonance branches execute on the physical GPU.
- CPU/GPU state results agree at sub-micrometre scale, and a separate Python sgp4 implementation agrees within 0.266 mm over 328 states.
- The full seven-day query reproduced 717 event identities with zero propagation errors and sub-0.1 ms maximum crossing-time difference against the packaged CPU event schedule.
- The direct kernel is compute-bound and register-limited rather than VRAM-bandwidth-bound; it sustained the stress run without immediate thermal instability.
- An event-buffer overflow was rejected explicitly instead of silently dropping events.

3.3 What the result does not prove

- It does not show where a real satellite was to sub-millimetre accuracy; those numbers measure implementation agreement under the same model.
- It does not include a complete TEME-to-ITRF/EOP, antenna, atmospheric, clock, covariance or RF-link stack.
- It does not make SGP4 seed data a truth ephemeris or an operational collision-avoidance product.
- It does not show large support or compatibility rejection: 99.245% of candidates survived support and compatibility rejected none of those survivors.
- It does not prove that direct reconstruction is always faster. The dense resident query is about 14.65x faster after the buffer exists.

Accurate product description

A GPU-native full-SGP4 analytical query substrate with a portable seed container and compact, identity-preserving pass events - not a compressed truth ephemeris and not a complete navigation stack.

4. Societal value mechanisms

The architecture creates public or economic value through mechanisms, not through the word “topological” by itself. Each mechanism below has a measurable benefit and a failure mode.

Value mechanism	How it works	Who benefits	Measure it with
Compact distribution	Ship model seeds, identities and updates rather than dense future state.	Low-connectivity stations, classrooms, field teams and edge devices.	Update bytes, source age, reconstruction coverage, integrity failures.
Local edge autonomy	Reconstruct and query without continuous cloud access.	Disaster sites, remote health reporting, industrial plants and	Offline uptime, energy, synchronization delay, missed

Value mechanism	How it works	Who benefits	Measure it with
		mobile robots.	events.
Event-first retention	Keep meaningful transitions and irreducible novelty instead of every sample.	Alerts, maintenance events, pass windows, threshold excursions.	Event yield, false/missed events, event ordering, legal retention needs.
Deterministic replay	Rebuild state from versioned seeds, code and novelty.	Audits, incident review, reproducible research and distributed operations.	Replay success, model/version provenance, signed manifests.
Analytic pruning	Reject by support and compatibility before expensive relation work.	Large catalogs, infrastructure graphs, public-health streams and transport networks.	SRG, CRG, false-negative rate and operator burden.
Finite guard semantics	Convert continuous state into explicit threshold crossings with margin and hysteresis.	Warnings, capacity transitions, safety envelopes and handovers.	Guard error ratio, chatter, lead time and alert precision.
Compute-storage choice	Trade arithmetic for storage and choose direct or cached paths per workload.	Memory-limited laptops, changing scenarios and one-shot screening.	First-query latency, repeated-query crossover, peak memory and energy.
Lineage and policy	Make identity, authority, route and external changes first-class.	Multi-agency alerts, maintenance history, digital twins and custody chains.	Provenance completeness, access-control failures, appeal/audit results.

Data minimization is a possibility, not an automatic property

Event-first storage can support purpose limitation and data minimization, but a seed may still reconstruct sensitive state. Privacy requires purpose, access control, retention rules, security and accountability. [R9; R16]

5. Application portfolio by sector

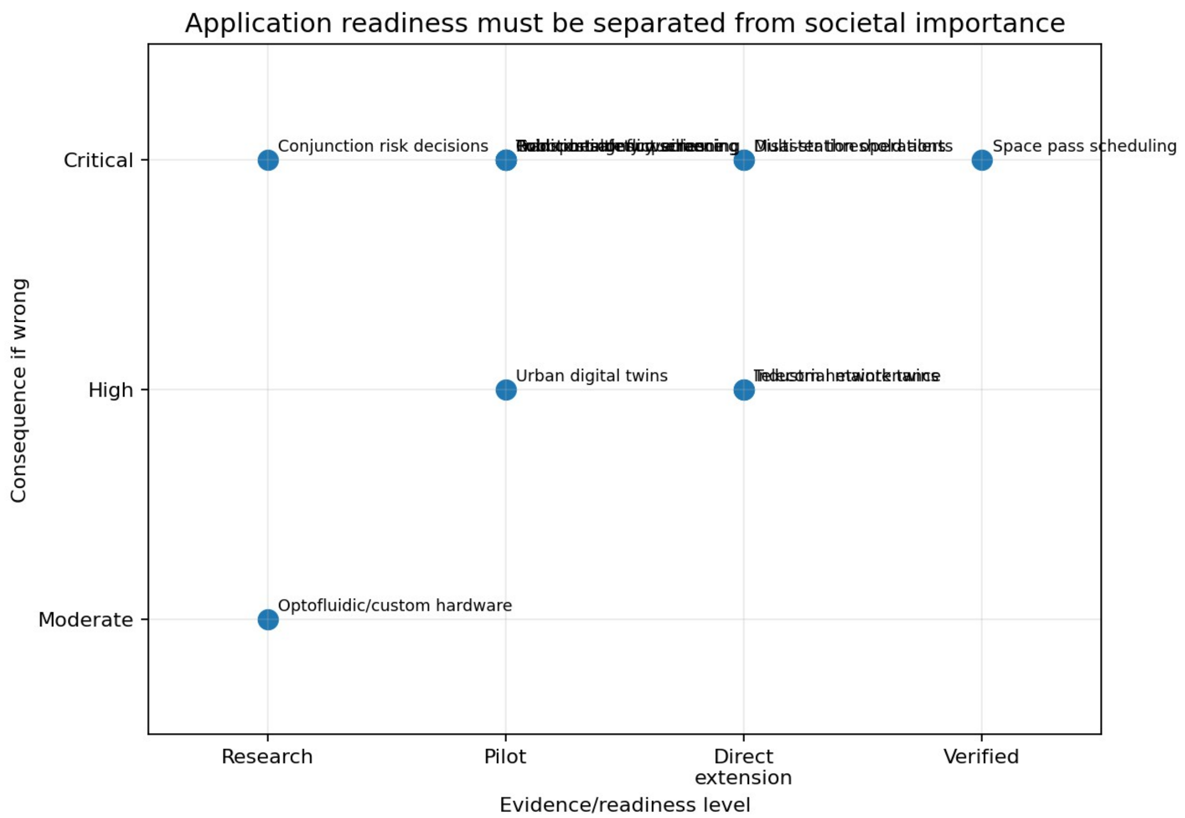

Figure 5. Societal importance and readiness are separate dimensions. High-consequence uses require stronger validation even when the mapping is technically natural.

Sector	Near-term opportunity	Boundary before operational use	Context
Space operations	Pass scheduling is verified; multi-station scheduling is a direct extension.	Conjunction and avoidance decisions need covariance, authoritative data and certification.	V0.5; R4-R6

Sector	Near-term opportunity	Boundary before operational use	Context
Disaster early warning	The support/guard/event pipeline closely matches end-to-end warning logic.	Hazard models, authoritative alerting, accessibility and false-alarm management are domain work.	R1-R3
Environment	Sparse change events can reduce imagery and sensor movement.	Detection models must retain uncertainty and raw evidence where required.	R3
Energy and utilities	Contingency and threshold screening are natural query-first workloads.	Protection/control actions remain safety-critical and utility-governed.	R13
Manufacturing	Seeded digital twins plus residual novelty fit predictive maintenance.	NIST emphasizes credibility, V&V and UQ before decision support is trusted.	R7-R9
Transport	Conflict windows, hazards and resource events fit support/compatibility/guard logic.	Safety certification, surveillance quality and interoperability are mandatory.	R12; R14
Public health	Large message streams can be reduced to investigation events with explicit policy.	Privacy, bias, jurisdiction, human review and uncertain ground truth dominate.	R10; R11; R18
Telecommunications	Network digital twins and handover/slice transitions are direct mappings.	Security and trust of network state are central.	R15; R9
Cities and buildings	Local digital twins can answer flood, heat, energy and accessibility queries.	Coverage gaps can reproduce social inequity; public governance matters.	R17; R14
Robotics	Motion seeds and finite safety guards can support local edge screening.	Sensor uncertainty and functional-safety certification remain decisive.	R7-R9

5.1 Space, communications and orbital services

Verified now: Generate local visibility/pass boundaries from portable orbital seeds; schedule recording, antenna time or operator attention.

Direct extension: Batch many stations or policy profiles without distributing dense trajectories.

Pilot: Use coarse support to pre-screen conjunction candidates before authoritative covariance-aware analysis.

Societal value: Lower entry cost for education, citizen science, remote stations, small missions and resilient communications planning.

Metrics: Element age, event identity, crossing time, station utilization, missed contacts, candidate throughput and per-query energy.

5.2 Disaster resilience and environmental response

Architecture fit: Hazard forecast or sensor state is reconstructed; affected areas form support; authority and recipient rules form compatibility; watch/warning thresholds are guards; alerts and route changes are events.

Practical outputs: CAP-compatible alert draft, route closure event, field-team task, map request or detailed-data fetch.

Societal value: Earlier focused warnings, lower field bandwidth and a smaller audit log for remote or overloaded operations.

Metrics: Lead time, false/missed alerts, population coverage, delivery latency, guard error, accessibility and operator burden.

Boundary: A model or GPU result cannot self-authorize a public warning. Official authority, calibrated hazard science and human procedures remain required.

5.3 Energy, infrastructure and industrial operations

Architecture fit: A validated asset or network model supplies expected state; telemetry becomes novelty; support localizes affected assets; a limit surface creates an event.

Practical outputs: Inspection event, maintenance work order, ranked grid contingency, microgrid transition request or process-safety alarm.

Societal value: Fewer unplanned outages, improved service continuity and lower data retention where full raw history is not mandatory.

Metrics: Downtime, lead time, precision/recall, service continuity, maintenance cost, energy/event and audit replay.

Boundary: Do not bypass certified protection or safety instrumented systems. The substrate should screen, explain and prioritize unless independently certified for control.

5.4 Transport, logistics and telecommunications

Architecture fit: Trajectory, route, service or network state can be reconstructed; support identifies intersecting domains; compatibility applies mode, capability and policy; guards produce conflicts, handovers or excursions.

Practical outputs: Conflict window, reroute event, port approach, handover, SLA breach or cold-chain quarantine event.

Societal value: Right information to the right operator, smaller distributed state and more explainable transitions.

Metrics: Conflict recall, p99 latency, service uptime, false alerts, resource utilization and bytes per update.

Boundary: Aviation, rail, telecom and food-safety operations have established standards and authorities that the substrate must integrate with rather than replace.

5.5 Public health and social services

Architecture fit: Seasonal and jurisdictional baselines are seeds; new messages are novelty; geography and population define support; data rights and case definitions define compatibility; anomaly thresholds create review events.

Practical outputs: Investigation queue, capacity escalation, outbreak signal or offline synchronization event.

Societal value: Faster triage of large message volumes and resilient operation in low-connectivity emergencies.

Metrics: Sensitivity, positive predictive value, alert burden, lead time, data completeness, coverage equity and privacy incidents.

Boundary: The event is a request for human investigation, not an automatic diagnosis or coercive decision.

6. Literal pragmatic deployment recipes

The following recipes convert the abstract pipeline into buildable pilot work. “Action” means an output to an authorized workflow; it does not imply autonomous control.

Recipe	Inputs	Compute	Guard	Action/output	Status
1. Ground-station scheduler	OMM CSV + station coordinates	Pack KSGP1; run one- or seven-day query	Elevation crosses mask	AOS/LOS calendar; antenna task	Verified
2. Multi-station allocator	KSGP1 + station fleet + calendars	Query all station/object pairs; rank conflicts	Usable contact window opens/closes	Allocated slot and conflict explanation	Direct extension
3. Flood warning node	Hydrologic seed + gauges + rainfall	Reconstruct local forecast; sync novelty	Water level crosses watch/warning	CAP alert draft and route task	Pilot
4. Transformer monitor	Thermal/aging model + telemetry	Predict hotspot and residual	Temperature or life margin crosses	Inspection/work order	Pilot
5. Machine maintenance	Digital twin + recipe + vibration	Rebuild expected state; retain residuals	Health score or RUL threshold	Maintenance event with lineage	Pilot
6. Health anomaly triage	Seasonal baseline + near-real-time messages	Aggregate by place/time/syndrome	Statistical guard crosses with persistence	Epidemiologist review event	Pilot
7. Aviation conflict screen	Flight plans + performance + weather	Intersect time-space supports	Separation/capacity margin crosses	Conflict event to SWIM-compatible service	Pilot
8. Telecom handover	Network twin + mobility + service	Reconstruct link/service state	Signal/QoS guard crosses	Handover or scale event	Direct extension

Recipe	Inputs	Compute	Guard	Action/output	Status
	policy				
9. Cold-chain custody	Route/thermal model + shipment identity	Integrate temperature-time dose	Dose or custody rule crosses	Quarantine/ inspection event	Direct extension
10. Robot keep-out	Kinematics + map + sensor updates	Predict reachable state	Distance/time-to-collision crosses	Stop/replan request	Pilot

6.1 Verified recipe: pass schedule in practice

1. Acquire only the orbital element set needed for the declared object group and respect the provider update/usage policy. [R4]
2. Pack identity-bearing seeds into KSGP1 and record source epoch, format, hash and model version.
3. Declare station latitude, longitude, altitude, elevation mask, time horizon and accepted model/frame limitations.
4. Run the direct GPU query. The verified seven-day case processed 19.35 million object-time intervals in 91.974 ms p50.
5. Use AOS/LOS events to schedule antenna or receiver work. Keep event identities, times and lineage; do not retain all positions unless needed.
6. Refresh elements according to the operational age policy and rerun. Treat a stale seed as an explicit fault, not a hidden approximation.
7. For operational radio use, add link budget, weather, antenna pattern, Doppler, clock and actual receiver validation.

6.2 Pilot recipe: event-first early warning

8. Choose one hazard and one authoritative threshold; avoid a multi-hazard platform as the first pilot.
9. Define the smallest reconstructible model and the external observations that must remain novelty.
10. Specify affected-area support and recipient/authority compatibility before computing a warning guard.
11. Use enter/exit hysteresis and a declared error margin; test tangencies, sensor gaps and delayed updates.
12. Emit a reviewable event with time, place, severity, source age, confidence and lineage.
13. Connect the output to an established alert protocol such as CAP only through the responsible authority. [R2]
14. Measure false/missed alerts, lead time, delivery success, population coverage and operator burden before scaling.

7. Metrics from kernel to societal outcome

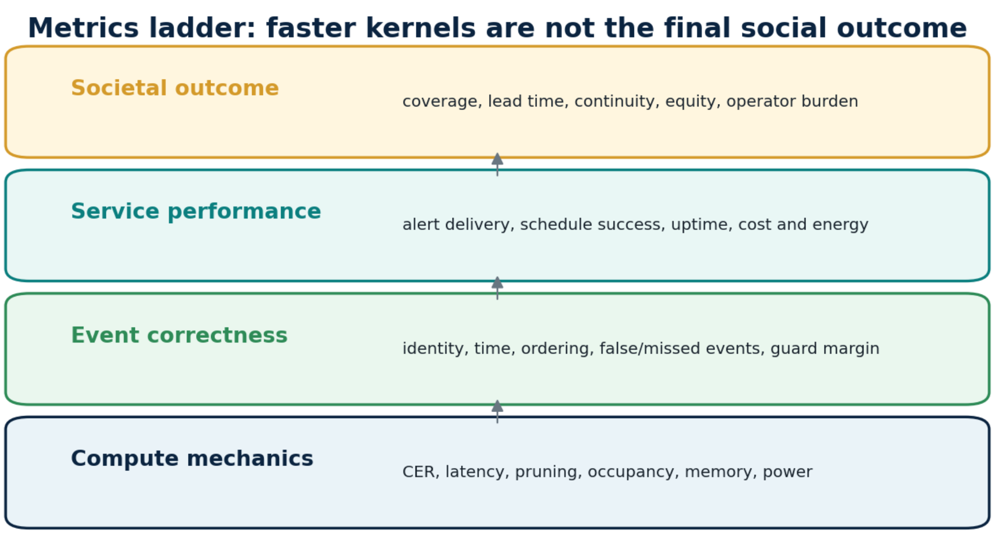

Figure 6. A faster kernel matters only if it preserves event correctness and improves service and societal outcomes.

The substrate already defines useful technical metrics such as candidate evaluation rate, support and compatibility rejection, event yield, state compression and commit cost. A societal deployment must add correctness, service and equity measures.

Layer	Representative metrics	Decision question
Compute	CER; p50/p95/p99; registers; occupancy; DRAM; power	Can the device sustain the workload?
Pruning	SRG; CRG; candidates reaching expensive work	Does query-first evaluation actually remove work?
Correctness	Guard error ratio; event identity/time/order; false/missed events	Does reduced state change the decision?
Persistence	Reconstructibility coverage; novelty rate; replay success	Can the state be rebuilt and audited?
Memory	Avoided materialization; peak footprint; source-to-container ratio	What storage/data movement is truly avoided?
Operations	Model age; uptime; overflow; alert delivery; operator burden	Will the service work in practice?
Society	Coverage; lead time; continuity; equity; cost and energy per useful event	Who benefits, who is missed and at what cost?

7.1 Verified SGP4 scorecard

Dimension	Assessment	Evidence
Numerical implementation	PASS	0 CPU/GPU method/error mismatches; independent maximum position difference 0.265627 mm.
Event correctness	PASS	717/717 identities; 0 propagation errors; maximum packaged-CPU crossing delta 0.000090514 s.
One-pass performance	PASS	91.974 ms direct versus 94.722 ms materialize-plus-query.
Repeated-query performance	CONDITIONAL	Resident dense query is 6.280 ms; cache wins after approximately the second identical query.
Memory avoidance	PASS WITH WORDING	5.8 KiB seeds avoid 590.6 MiB of chosen double4 positions; ratio is horizon-dependent.
Support pruning	WEAK IN THIS TEST	Only 0.755% of candidates rejected; compatibility rejected none.
GPU bottleneck	KNOWN	146 registers/thread, 23.72% occupancy, 0.008% DRAM utilization.
Thermal feasibility	PASS FOR 98 s	70 C maximum; 73.1 W average active power; approximately 0.48% settled slowdown.
Operational navigation	NOT READY	Full EOP/frame, RF, covariance, uncertainty and certification absent.

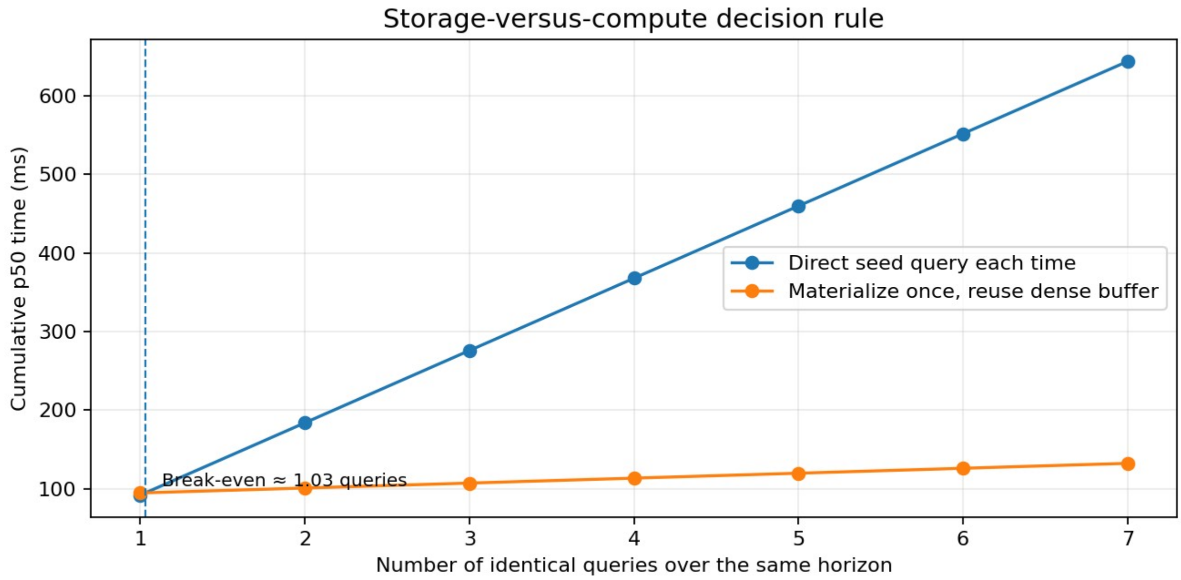

Figure 7. The direct path is a one-shot advantage; repeated identical queries favor materialization and reuse.

7.2 Application acceptance gates

- Guard error remains comfortably below the declared event margin across representative and adversarial cases.
- False accepts, false rejects, missed events and event-order changes stay within the application budget.
- Support and compatibility remove enough work to justify the architecture, or direct fusion avoids enough materialization to compensate.
- Event/FIFO compaction does not lose data under worst-case event density.
- The model freshness policy, source provenance and external novelty path are explicit.
- Full-system energy and operating cost are measured rather than inferred from memory reduction.
- A conventional dense or database solution is not cheaper, faster or more accurate at equal error.

8. Governance, privacy, safety and trust

A societal substrate is a governance system as much as a compute system. The compatibility predicate can encode authority, purpose and access, while lineage can make transitions auditable. Those features are valuable only when they are explicit, reviewable and secured.

Governance question	Risk	Required control
Model authority	Who owns the model, threshold and update cadence?	Signed model/seed manifest; source-age policy; independent reference.
Data purpose	Why is this state reconstructed and who may query it?	Purpose-bound query profile; access control; data-minimization assessment. [R16]
Uncertainty	Can the guard be trusted near the boundary?	UQ, calibrated confidence, error budget and abstention. [R8]
Human oversight	Who approves high-consequence events or actions?	Review queues, escalation and explicit non-autonomous modes.
Lineage security	Can identities and logs be forged or reordered?	Cryptographic signatures, monotonic sequence, secure time and append-only storage. [R9]
Equity	Do support, compatibility or thresholds exclude groups or regions?	Coverage/error audit by geography and affected population.
Interoperability	Can downstream authorities consume the event?	Versioned schema and established standards such as CAP, OMM or SWIM. [R2; R4; R12]
Fail-safe behavior	What happens when data are stale, buffers overflow or models disagree?	Explicit fault events, no silent truncation, conservative fallback and operator notice.

Lineage is not identity proof

The compact 32-bit lineage hash described by UGTS is a routing and validation checksum. It is not a cryptographic identity, signature or durable provenance system. Production deployments need signed manifests, secure update channels and an authenticated event log. [UGTS p. 10; R9]

8.1 Domain-specific red lines

Domain	Red line
Emergency alerts	Do not issue public warnings without recognized authority, calibrated hazard models and accessible dissemination.
Health	Do not treat anomaly events as diagnoses; protect personal data and test bias/coverage.
Grid/process control	Do not bypass certified protection or safety systems.
Aviation/robotics	Do not promote screening kernels to safety authority without certification and sensor uncertainty.
Space safety	Do not use SGP4-only geometry as a collision manoeuvre decision; use covariance-aware authoritative services.
Public administration	Do not automate rights-affecting decisions solely through a compatibility mask or guard.

9. Economics and direct-query versus cache decisions

The verified application exposes the central economic trade: arithmetic can replace persistent derived state, but only for the right query pattern.

Pattern	Use it when	Main cost/risk
Direct reconstruction	One-shot or changing horizon/criteria; memory- or bandwidth-constrained edge; compact distribution.	Repeated full-model compute, high register pressure and model-validation cost.
Dense materialization	Many repeated queries over one stable horizon; enough VRAM; simple downstream analytics.	Large memory, data movement and rebuild cost when horizon or source changes.
Hybrid cache	Frequently queried windows or selected objects; cache only hot derived state.	More complex invalidation, provenance and consistency logic.
Event-only retention	Sparse, reconstructible dynamics and legally sufficient event record.	Cannot be used when raw samples are authoritative evidence or reconstruction is uncertain.

Verified break-even: Q = materialization / (direct - dense query) = 88.443 / (91.974 - 6.280) ≈ 1.03. One query favors direct; from the second identical query onward, dense reuse is faster in this workload.

Storage wording: The KSGP1 container is 5,793 bytes and the chosen seven-day double4 position buffer is 619,316,224 bytes. The correct statement is that the seeds avoid materializing that derived buffer. KSGP1 is actually larger than the 4,852-byte source OMM CSV because it adds typed records, timeline metadata and integrity structure.

Economic kill criterion

Promote the substrate only when the complete service - source acquisition, model updates, compute, events, storage, communications, operator time, calibration and failures - is better than a conventional reference at equal accuracy and safety. [UGTS p. 15]

10. Recommended pilot roadmap

The next work should preserve the strongest verified asset - deterministic seed-to-event GPU execution - while adding domain truth, stronger pruning and operational governance.

Time	Objective	Work	Deliverable
0-90 days	Harden the verified space application	Fix build-script exit handling; canonicalize CSV newlines; add	Reproducible one-command build and a public-safe multi-

Time	Objective	Work	Deliverable
		live OMM refresh with source-age rules; multi-station queries; signed manifests; 30-60 minute endurance.	station demo.
3-6 months	Raise orbital accuracy and utility	Add UT1/EOP and polar motion; validate TEME-to-ITRF; station masks, Doppler and link-budget plugin; uncertainty and element-age policy; specialize branches to reduce registers.	Ground-station planning service with bounded accuracy and resource scheduling.
3-6 months	Run one non-space societal pilot	Choose a clear threshold domain such as machine health or flood gauges; define independent truth, guard margins, false/missed events and operator workflow.	Evidence that the architecture generalizes beyond orbit propagation.
6-12 months	Scale catalog and pruning	Mixed-orbit catalog, thousands of objects, multiple stations, support indexing, compatibility policies and event-density stress.	Measured SRG/CRG advantage rather than mainly materialization avoidance.
6-12 months	Operational trust layer	Cryptographic provenance, access control, policy versioning, audit replay, secure update path and failure injection.	Deployable event substrate component rather than only benchmark code.
Research	High-consequence and hardware endpoints	Covariance-aware conjunction pipeline, safety certification, FPGA/ASIC gate or optofluidic experiments with full-system energy and calibration.	Separate research programmes with explicit go/no-go gates.

10.1 Recommended next three pilots

Pilot	Why this one	Primary evidence to collect
1. Multi-station satellite scheduler	Closest to the verified code; creates real utility without claiming navigation or collision authority.	Contacts scheduled, conflicts resolved, update age, latency, utilization and missed passes.
2. Industrial machine-health event log	Strong fit for reconstructible expected dynamics plus sparse residual novelty; clear business and audit metrics.	Lead time, precision/recall, downtime, novelty density, maintenance cost and replay.
3. Flood/river threshold edge node	High public value and a natural guard/event pattern; tests low-connectivity and authoritative alert integration.	Lead time, false/missed alerts, coverage, delivery, energy and human response.

11. Conclusions

Societal conclusion

The verified SGP4 deployment demonstrates a useful and general architectural proposition: compact identity-bearing seeds can be reconstructed on demand, screened by explicit support and guard logic, and reduced to a small, auditable event stream on commodity GPU hardware. That is a real enabling capability for edge analytics, digital twins, early-warning systems and high-volume screening.

The central value is not “infinite compression” or exotic topology. It is disciplined separation of reconstructible state from irreducible novelty; local relevance from compatibility; continuous trajectories from meaningful transitions; and current coordinates from persistent identity. The same separation is useful in space operations, infrastructure, manufacturing, health and emergency management when a trustworthy domain model exists.

The central risk is equally clear: a compact model can be confidently wrong. Societal deployment therefore depends on model authority, uncertainty, event margins, false/missed-event budgets, secure lineage, equitable coverage, human oversight and comparison with a conventional baseline. The UGTS kill criteria are not a disclaimer; they are the promotion gate.

The most defensible current claim is bounded: the query-first seed-to-event mechanism works on the stated laptop for the verified full-SGP4 analytical application. The broader societal portfolio is a structured programme of pilots, not a claim that all applications are already solved.

Appendix A. Comprehensive application matrix

The complete machine-readable version is included as UGTS_Application_Matrix.csv. This appendix shows the operational core of each mapping.

ID	Sector	Use case	Readiness	Seed / novelty	Support / compatibility	Guard / output	Metrics / caveat
A01	Space and communications	Local ground-station pass scheduling	Verified application	Seed: OMM/GP mean elements, station location and query policy Novelty: Fresh element-set updates, station outages, antenna masks	Support: Object and time inside declared data horizon Compatibility: Station, object class, antenna/radio capability and policy match	Guard/event: Elevation crosses the configured horizon mask; emit AOS or LOS Output: Schedule receiver, antenna, recording or operator task; retain event identity and time	Metrics: Crossing-time error, event identity match, data age, candidates/s, first-query latency, missed/false passes Caveat: Current deployment is not a full RF link budget or navigation system.
A02	Space and communications	Multi-station antenna and downlink resource scheduling	Direct extension	Seed: Same orbital seeds plus station fleet and resource calendars Novelty: Weather, maintenance, frequency conflicts, operator priorities	Support: Pass intersects station visibility and time window Compatibility: Band, antenna, modulation, licensing and mission priority	Guard/event: Start/end of usable contact or conflict window Output: Allocate contact slot and produce auditable conflict explanation	Metrics: Scheduled contacts, conflict rate, utilization, missed contacts, solve latency, fairness Caveat: Needs link budgets, regulatory constraints and weather models.
A03	Space safety	Conjunction pre-screening before high-fidelity collision analysis	Pilot candidate	Seed: Orbital state models and object identities Novelty: New tracking observations, covariance updates, manoeuvre notices	Support: Spatial/time encounter envelope Compatibility: Object classes, manoeuvrability and screening policy	Guard/event: Predicted separation crosses a coarse threshold Output: Forward only candidates to authoritative conjunction assessment	Metrics: Recall, false-negative rate, screening reduction, time-to-alert, covariance-aware miss distance Caveat: SGP4-only positions are insufficient for operational collision decisions.
A04	Disaster risk reduction	Flood and river-level threshold warning	Pilot candidate	Seed: Catchment topology, rating curves, forecast model and sensor calibration Novelty: Rainfall, gauge readings, obstructions and model corrections	Support: Affected basin, community and forecast horizon Compatibility: Alert authority, hazard type, language and recipient policy	Guard/event: Forecast or measured level crosses watch, warning or evacuation threshold Output: Emit an authoritative CAP-compatible alert draft and route to responsible officials	Metrics: Lead time, false/missed alerts, guard error, coverage, alert delivery latency, model age Caveat: Requires calibrated hydrology and official warning authority; model output is not itself an alert.
A05	Disaster risk reduction	Wildfire perimeter and evacuation-route crossing	Pilot candidate	Seed: Road graph, terrain, vegetation and evacuation plans Novelty: Observed fire perimeter, wind, closures and resource status	Support: Routes and communities within a hazard cone or travel-time region Compatibility: Vehicle class, responder role and route policy	Guard/event: Predicted hazard reaches route, facility or evacuation-time margin Output: Flag route loss, reroute decision or resource staging event	Metrics: Route-availability lead time, missed closures, reroute latency, false alarm burden Caveat: Fire behavior is uncertain and needs authoritative models and human command.
A06	Environment and climate	Satellite-derived change-event extraction	Direct extension	Seed: Baseline geometry, sensor model and prior scene state Novelty: New observations and irreducible detected changes	Support: Geographic and temporal area of interest Compatibility: Sensor, resolution, policy and confidence class	Guard/event: Change score crosses a calibrated threshold with persistence/hysteresis Output: Create a compact change event and request detailed imagery only where needed	Metrics: Detection recall, false-change rate, bytes transferred, latency, confidence calibration Caveat: Image reconstruction and change detection models must be independently validated.
A07	Environment and utilities	Water-quality, pressure and leak-event monitoring	Pilot candidate	Seed: Network topology, hydraulic model, normal operating envelope Novelty: Sensor residuals, valve changes, repairs and contamination observations	Support: Hydraulic zone and time window affected by an anomaly Compatibility: Sensor quality, authority and contaminant/asset class	Guard/event: Pressure, flow or quality residual crosses a validated threshold Output: Create investigation event; isolate zone only through certified controls	Metrics: Detection delay, leak localization, false alarms, retained bytes, sensor coverage Caveat: A compact event log cannot replace legally required raw evidence or calibrated instrumentation.
A08	Energy and utilities	Grid voltage, frequency and thermal-limit screening	Pilot candidate	Seed: Network model, asset ratings and operating state Novelty: Telemetry, outages, weather, switching and distributed-resource updates	Support: Affected network region and contingency horizon Compatibility: Protection scheme, operating mode and operator policy	Guard/event: Voltage, frequency, loading or stability margin crosses limit Output: Rank violations and propose, not autonomously execute, corrective actions	Metrics: Contingencies/s, recall, false alarms, margin error, operator workload, time-to-decision Caveat: Power-system models and protection actions require

ID	Sector	Use case	Readiness	Seed / novelty	Support / compatibility	Guard / output	Metrics / caveat
							utility-grade validation and safety governance.
A09	Energy and utilities	Microgrid islanding and reconnection events	Pilot candidate	Seed: Microgrid topology, resource capacities and controller state Novelty: Load, generation, storage, fault and weather updates	Support: Local island and connection point Compatibility: Grid code, protection state, resource availability	Guard/event: Frequency/voltage stability and synchronization guard crosses Output: Request operator/controller transition with signed event record	Metrics: Continuity minutes, transition success, false trips, energy/event, recovery time Caveat: Automatic switching is safety-critical and must use certified controls.
A10	Manufacturing	Predictive maintenance and machine-health events	Strong pilot candidate	Seed: Machine model, operating recipe and maintenance history Novelty: Vibration, temperature, load and repair residuals	Support: Operating regime and component affected Compatibility: Machine configuration, product batch and sensor validity	Guard/event: Residual or predicted remaining-life margin crosses threshold Output: Create work order or inspection event with lineage to machine and batch	Metrics: Lead time, precision/recall, downtime avoided, novelty rate, maintenance cost, guard margin Caveat: Credibility requires verification, validation and uncertainty quantification.
A11	Manufacturing and process safety	Process safety-envelope monitoring	Pilot candidate	Seed: Validated process model, equipment limits and recipe Novelty: Measurements, material properties, operator changes	Support: Current process phase and affected equipment Compatibility: Recipe, safety state and authorization	Guard/event: Pressure, temperature, composition or rate crosses a multi-variable safety surface Output: Alarm, safe-state request or operator confirmation; retain full audit lineage	Metrics: Missed-event probability, nuisance alarms, response time, event order, safety integrity Caveat: Must not bypass certified safety instrumented systems.
A12	Transport and aviation	Airspace conflict-window and flow screening	Pilot candidate	Seed: Flight plans, performance models, airspace geometry Novelty: Weather, reroutes, delays and surveillance updates	Support: Aircraft pairs and sectors with intersecting time-space envelopes Compatibility: Altitude, rules, wake category and operational policy	Guard/event: Separation or capacity margin crosses threshold Output: Publish conflict or flow-management event to interoperable services	Metrics: Conflict recall, false alerts, p95 latency, sector reduction, update age Caveat: Operational use requires certified surveillance, uncertainty and aviation safety processes.
A13	Transport infrastructure	Road, rail, port and pipeline hazard windows	Pilot candidate	Seed: Asset graph, geometry, inspection model and service rules Novelty: LiDAR, sensor, weather, work-zone and incident updates	Support: Assets within hazard or degradation domain Compatibility: Asset class, vehicle/service and maintenance policy	Guard/event: Condition or hazard crosses operational threshold Output: Inspection, restriction, reroute or maintenance event	Metrics: Hazards detected, inspection reduction, false closures, service continuity, response time Caveat: Digital models must be kept current and must not hide uncertainty.
A14	Logistics and food safety	Cold-chain excursion and custody events	Direct extension	Seed: Route plan, thermal model, packaging specification and shipment identity Novelty: Temperature, door, delay and custody changes	Support: Shipment segment and time range Compatibility: Product class, regulatory limit and sensor trust	Guard/event: Temperature-time dose or custody policy crosses threshold Output: Quarantine, inspect or accept with auditable event chain	Metrics: Excursion duration, false alarms, spoilage, event completeness, retained data volume Caveat: Some regulations require raw sensor retention; event-only storage may be insufficient.
A15	Public health	Syndromic anomaly and investigation triggers	Pilot candidate	Seed: Seasonal baselines, syndrome definitions and jurisdictional model Novelty: Emergency visits, laboratory, mortality and environmental data	Support: Place, population and time window Compatibility: Data-use rights, syndrome definition, facility quality and policy	Guard/event: Observed count or residual crosses an alert threshold with persistence Output: Create epidemiologist review event, never an automatic diagnosis	Metrics: Lead time, sensitivity, positive predictive value, alert burden, equity coverage, privacy Caveat: Health use requires strong governance, de-identification, bias assessment and human review.
A16	Public health emergencies	Offline outbreak reporting and event synchronization	Direct extension	Seed: Case definitions, facility list, reporting schedule and local baseline Novelty: New reports, unusual clusters and connectivity restoration	Support: Affected place and reporting period Compatibility: Authorized reporter, case definition and data-quality rules	Guard/event: Case count, cluster or severity criterion crosses Output: Queue signed alert/investigation event for local and central teams	Metrics: Offline uptime, synchronization delay, missing reports, false/missed alerts, device energy Caveat: The current code is not a health system; this is an architectural mapping.
A17	Healthcare operations	Hospital capacity and ambulance-diversion thresholds	Pilot candidate	Seed: Capacity model, staffing plan and service rules Novelty: Admissions,	Support: Facility, service line and forecast horizon Compatibility: Patient/service	Guard/event: Bed, staff, queue or equipment margin crosses Output: Escalation, diversion	Metrics: Warning lead time, diversion precision, queue time, coverage and fairness

ID	Sector	Use case	Readiness	Seed / novelty	Support / compatibility	Guard / output	Metrics / caveat
				discharges, staffing and equipment changes	category, policy and data authorization	recommendation or mutual-aid request	Caveat: Operational decisions remain clinical and administrative responsibilities.
A18	Telecommunications	Network handover, slice and service-degradation events	Direct extension	Seed: Topology, radio/transport model, service policies and slice definitions Novelty: Traffic, faults, mobility and configuration updates	Support: Users/flows in affected cell, link or service region Compatibility: Slice, QoS, authorization and device capability	Guard/event: Signal, latency, loss, utilization or policy margin crosses Output: Handover, reroute, scale or operator alert event	Metrics: Outage minutes, handover failures, SLA violations, p99 latency, events/J Caveat: Network digital twins introduce cybersecurity and trust risks.
A19	Cities and buildings	Urban flood, heat and building-energy event queries	Pilot candidate	Seed: Building/road geometry, energy and hazard models Novelty: Weather, occupancy, sensor and construction updates	Support: Neighbourhood, asset and time horizon Compatibility: Service authority, vulnerability class and intervention policy	Guard/event: Flood depth, heat exposure, energy or accessibility threshold crosses Output: Inspection, cooling centre, road closure or retrofit-priority event	Metrics: Population coverage, lead time, false alerts, energy demand, equity distribution Caveat: Model coverage and public participation determine whether benefits are equitable.
A20	Robotics and autonomy	Keep-out, collision and task-transition events	Pilot candidate	Seed: Robot kinematics, motion primitives and map Novelty: Detected obstacles, human positions, task changes and calibration	Support: Reachable space and time horizon Compatibility: Robot mode, payload, human-safety zone and policy	Guard/event: Distance, time-to-collision or stability margin crosses Output: Stop, replan or request human intervention	Metrics: Missed collisions, false stops, reaction latency, guard error, energy/event Caveat: Safety certification and sensor uncertainty are mandatory.
A21	Cyber-physical security	Policy-compatible anomaly and control-transition log	Research candidate	Seed: Authorized system model, identities and policy graph Novelty: Configuration, telemetry, alerts and operator actions	Support: Assets and time ranges reachable from the anomaly Compatibility: Identity, role, policy and trust state	Guard/event: Behavior or control state crosses a validated security boundary Output: Investigation, containment proposal and signed evidence record	Metrics: Detection recall, false positives, replay success, provenance completeness, containment time Caveat: Compact lineage hashes are not cryptographic proof and must be backed by signatures and secure logs.
A22	Science and simulation	Sparse event retention for large deterministic simulations	Direct extension	Seed: Model version, initial conditions and deterministic parameters Novelty: External forcing, interventions and irreducible observations	Support: Requested region, entity and time window Compatibility: Model version, units and scenario policy	Guard/event: Scientific threshold, contact, phase transition or failure surface crosses Output: Retain event lineage and reconstruct detail on demand	Metrics: Reconstruction error, novelty density, storage avoided, replay time, reproducibility Caveat: Chaotic or stochastic models may require more checkpoints and uncertainty state.
A23	Public administration	Auditable eligibility or permit-state transition support	Research candidate	Seed: Versioned rules, application facts and authority state Novelty: New evidence, appeals, policy changes and human decisions	Support: Relevant jurisdiction and case type Compatibility: Legal authority, role and purpose	Guard/event: A rule-defined status transition becomes valid Output: Draft an explainable case event for human determination	Metrics: Decision consistency, appeal rate, processing time, explanation completeness, disparate impact Caveat: Do not automate rights-affecting decisions without legal, fairness and human-review safeguards.
A24	Education and citizen science	Portable deterministic experiments on consumer GPUs	Verified enabling capability	Seed: Open model, seeds, station or sensor parameters Novelty: Student scenarios and measured observations	Support: Declared experiment domain Compatibility: Version, units and task settings	Guard/event: Experiment-specific threshold crossing Output: Reproducible result package and compact event log	Metrics: Reproduction success, hardware reach, package size, runtime, learning outcomes Caveat: Educational reproducibility is not operational certification.

Appendix B. Metric dictionary

Symbol	Name	Formula	Purpose	Required context
CER	Candidate Evaluation Rate	N / device_time	Raw analytical screening throughput	device, batch, profile, precision
p50/p95/p99	Query latency percentiles	distribution of end-to-end latency	Typical and tail response time	same workload and warm-up policy
SRG	Support Rejection Gain	N / N_support	How much local support prunes before expensive work	support definition and false-negative budget
CRG	Compatibility Rejection Gain	N_support / N_compatible	How much policy/mode/schema compatibility prunes	compatibility schema and fairness review
EY	Event Yield	N_events / N_candidates	Event sparsity and compaction opportunity	event definition and time resolution
VET	Verified Event Throughput	N_verified / device_time	Useful event output rate	support, compatibility, guard and confidence
AMR	Avoided Materialization Ratio	dense derived bytes / seed package bytes	Derived state that need not be stored	horizon, fields and sample interval
SCR	Source-to-container Ratio	source bytes / container bytes	Whether the container is smaller than its source input	same semantic content
NER	Novelty Event Rate	novelty records / state updates	How much state is truly irreducible	novelty definition and checkpoint policy
GER	Guard Error Ratio	max guard error / declared event margin	Whether numeric error can alter decisions	validated reference and margin
EIMR	Event Identity Match Rate	matched event identities / reference events	End-to-end correctness of who/what crossed	independent reference and identity schema
ETE	Event-Time Error	|t_predicted - t_reference|	Timing accuracy of transitions	time standard and reference
FA/FR/MR	False accept / false reject / miss rates	confusion matrix rates	Operational trust and safety	ground truth and sample representativeness
EOM	Event Order Mismatches	count of ordering differences	Whether transitions remain causally consistent	stable identity and timestamp policy
MFA	Model Freshness Age	now - source epoch/update time	Risk from stale seeds and state	domain-specific maximum age
RC	Reconstructibility Coverage	rebuildable state / total required state	What fraction can be regenerated safely	model family and error budget
RQX	Repeated Query Crossover	Q where direct total = materialize + Q*dense_query	When caching becomes faster	same horizon and criteria
PMF	Peak Memory Footprint	maximum host/device bytes	Hardware feasibility	all buffers, caches and logs
EPVE	Energy per Verified Event	full-system joules / verified events	Energy efficiency of useful output	source, GPU, control and communication energy
TDS	Thermal Drift Slowdown	(settled p50 - cold p50) / cold p50	Sustained laptop stability	duration, power mode and ambient conditions
RO	Register and Occupancy Profile	registers/thread and achieved occupancy	GPU bottleneck classification	kernel, architecture and compiler
DUT	Data Update Traffic	bytes per authoritative update	Network/storage burden	refresh cadence and source policy
FIFOO	FIFO/Buffer Overflow Rate	lost or rejected events / produced events	Whether compaction loses events	buffer capacity and backpressure
ARS	Audit Replay Success	identical/reconciled outputs / replay attempts	Deterministic provenance and incident review	versioned seeds, code and novelty log
COV	Population / Asset Coverage	served units / target units	Societal reach	denominator and inclusion criteria
ALT	Action Lead Time	actionable event time - hazard/need onset	Practical warning or response advantage	authoritative reference
FAB	False Alarm Burden	false alerts per operator/community/time	Human workload and trust	alert severity and recipient group
EQC	Equity Coverage	coverage and error by affected group/region	Distribution of benefit and harm	legitimate demographic/geographic audit
SCT	Service Continuity	minutes/hours of maintained service	Resilience benefit	baseline and incident definition

Appendix C. Verified SGP4 metrics and evidence inventory

Metric	Value	Scope	Source
Device	NVIDIA GeForce RTX 5070 Ti Laptop GPU	Physical laptop run	V0.5
Compute capability	12.0 / sm_120	Native CUBIN plus PTX present	V0.5
VRAM reported	12,227 MiB	Device inventory	V0.5
Objects	32 GPS operational objects	Seven-day application	V0.5
Horizon	604,800 s (7 days)	One-second intervals	V0.5
Candidate intervals	19,353,600	Object-time interval evaluations	V0.5
Direct query p50	91.974 ms	Full SGP4 + guard query	V0.5
Direct query p95	92.000 ms	Tail latency	V0.5
Direct throughput	210.4 M candidates/s	Seven-day run	V0.5
1 GiB stress throughput	207.7 M candidates/s	Stress horizon	V0.5
2 GiB stress throughput	212.2 M candidates/s	Stress horizon	V0.5
Dense materialization p50	88.443 ms	Build double4 position buffer	V0.5
Resident dense query p50	6.280 ms	Query after materialization	V0.5
Dense end-to-end p50	94.722 ms	Materialize + one query	V0.5
First-query speedup	1.030x	Direct vs dense end-to-end	V0.5
Resident dense speed advantage	14.65x	Repeated identical horizon	V0.5
KSGP1 container	5,793 bytes	Portable seeds + metadata	V0.5
Avoided double4 materialization	619,316,224 bytes (590.6 MiB)	Seven-day position-only buffer	V0.5
Avoided-materialization ratio	106,907.69x	Horizon-dependent; not ordinary byte compression	V0.5
Supported intervals	19,207,536	0.755% rejected by support	V0.5
Compatibility survivors	19,207,536	No extra rejection in this workload	V0.5
Visible endpoints	5,498,429	Ground-station elevation test	V0.5
AOS / LOS events	359 / 358	Total 717 compact events	V0.5
Propagation errors	0	Seven-day run	V0.5
Event identity match	717 / 717 exact	GPU direct vs packaged CPU events	V0.5
Maximum crossing-time delta	0.000090514 s	GPU direct vs packaged CPU event schedule	V0.5
CPU/GPU max position delta	0.000383791 mm	256 GPS validation states	V0.5
Independent max position delta	0.265627 mm	Python sgp4 2.25 / WGS72, 328 states	V0.5
Sanitizer findings	0 memcheck errors; 0 race hazards; 0 sync errors	Adversarial runtime checks	V0.5
Forced overflow behavior	Explicit failure; no silent truncation	11 events into capacity 1	V0.5
Direct kernel registers	146 registers/thread	No ptxas spills reported	V0.5
Direct achieved occupancy	23.72%	Register-limited	V0.5
Direct DRAM utilization	0.008%	Compute-bound, not memory-bound	V0.5
Sustained duration	about 98.4 s	600-repeat direct-only run	V0.5
Sustained p50	162.328 ms	Thermally settled 1 GiB workload	V0.5
Maximum temperature	70 C	Sustained run	V0.5
Average active GPU power	73.1 W	Sustained run	V0.5
Peak sustained power	74.8 W	Sustained run	V0.5
Average active graphics clock	2,763 MHz	Sustained run	V0.5
Maximum observed VRAM	2,752 MiB	Seed/direct workload	V0.5

Appendix D. References and provenance

Web references were accessed on 16 August 2026. Third-party pages are referenced but not redistributed. Recommendations are engineering inferences unless explicitly marked as verified.

[UGTS] Unified Geometric-Topological Substrate, UGTS-GN 1.1: GPU-native compilation, physical-device architecture, performance, memory and compression addendum. User-supplied source document (2026). Primary architecture source. File SHA-256: cb95f7e9036e1b89e009c88ae1849ee71cddb3bdb6629c98593e32f3d65254f8.

[V0.5] KLB SeedChain GPU v0.5.0 - Full SGP4/SDP4 RTX 5070 Ti Laptop verification archive. User-supplied measured deployment evidence (2026). Primary measured evidence. Source archive SHA-256: 3f9590144b71b79f9fb698f3dd0c1ba22f127a66c09ccab7cae364257cd304bb.

[R1] Advisory Panel Joint Statement of the Early Warnings for All Initiative and Priority Actions. United Nations Office for Disaster Risk Reduction (UNDRR) (2025). https://www.undrr.org/publication/documents-and-publications/advisory-panel-joint-statement-early-warnings-all-initiative. Calls for data-driven, user-tailored and people-centred multi-hazard warning systems.

[R2] Common Alerting Protocol (CAP). World Meteorological Organization (WMO) (2024-2026). https://wmo.int/activities/common-alerting-protocol-cap. Standardized multi-hazard alert format and authoritative-source registry.

[R3] About Copernicus Emergency Management Service On-Demand Mapping. European Commission / Copernicus EMS (2025). https://mapping.emergency.copernicus.eu/about/. Satellite and geospatial products for preparedness, rapid response and recovery.

[R4] A New Way to Obtain GP Data (OMM/TLE formats and SGP4 semantics). CelesTrak (2020; updated 2026). https://celestrak.org/NORAD/documentation/gp-data-formats.php. Describes OMM-compatible GP data, TEME/UTC/SGP4 fields and current query formats.

[R5] Reentry and collision avoidance. European Space Agency (ESA) Space Safety (Current service description). https://www.esa.int/content/view/full/413425. Operational conjunction analysis depends on orbit knowledge, geometry and uncertainty.

[R6] CREAM: avoiding collisions in space through automation. European Space Agency (ESA) Space Safety (2025). https://www.esa.int/Space_Safety/Space_Debris/CREAM_avoiding_collisions_in_space_through_automation. Automation aims to reduce operator workload, false alerts and response time.

[R7] Digital twins. National Institute of Standards and Technology (NIST) (Current programme page). https://www.nist.gov/digital-twins. Digital twins support monitoring, forecasting, diagnosis, optimization and decision support.

[R8] Credibility Consideration for Digital Twins in Manufacturing. NIST (2022). https://www.nist.gov/publications/credibility-consideration-digital-twins-manufacturing. Verification, validation and uncertainty quantification are required for trustworthy decisions.

[R9] Security and Trust Considerations for Digital Twin Technology (NIST IR 8356). NIST (2025). https://www.nist.gov/publications/security-and-trust-considerations-digital-twin-technology-digital-twin-technology. Covers cybersecurity and trust for real-time monitoring, control and simulation.

[R10] About the National Syndromic Surveillance Program (NSSP). U.S. Centers for Disease Control and Prevention (CDC) (2026). https://www.cdc.gov/nssp/php/about/index.html. Near-real-time integrated health data supports early detection and response.

[R11] Early Warning, Alert and Response System (EWARS). World Health Organization (WHO) (Current programme page). https://www.who.int/emergencies/surveillance/early-warning-alert-and-response-system-ewars/. Rapid disease-surveillance deployment in emergency, conflict and low-connectivity settings.

[R12] System Wide Information Management (SWIM) Program Overview. Federal Aviation Administration (FAA) (2025). https://www.faa.gov/air_traffic/technology/swim/overview. Aviation data-sharing backbone intended to deliver the right information to the right users at the right time.

[R13] Grid Modernization Initiative. U.S. Department of Energy (Current programme page). https://www.energy.gov/gmi/grid-modernization-initiative. Grid tools are developed to measure, analyze, predict, protect and control future power systems.

[R14] INSIGHTS Project: continuously updated digital models of transportation infrastructure. U.S. Department of Transportation (2025). https://www.transportation.gov/research-and-technology/insights-project. Digital models of roads, ports, rail, airports and pipelines for safety, maintenance and evacuation planning.

[R15] Digital twin network standards (Y.3090, Y.3093 and Y.3168). International Telecommunication Union (ITU-T) (2022-2025). https://www.itu.int/rec/T-REC-Y.3090/en. Frameworks for network digital twins, data domains and network-slice applications.

[R16] Regulation (EU) 2016/679, Article 5. EUR-Lex / European Union (2016; current consolidated text). https://eur-lex.europa.eu/eli/reg/2016/679/oj?locale=EN. Includes purpose limitation, data minimisation, accuracy, storage limitation, security and accountability.

[R17] EDIC Local Digital Twins and CitiVERSE. European Commission, Regional and Urban Policy (Current programme page). https://regions-and-cities.ec.europa.eu/cities-portal/eu-initiatives-cities/european-digital-infrastructure-consortium-edic-local-digital-twins-and-citiverse_en. Real-time digital replicas of urban assets and processes for planning and service management.

[R18] Epidemic Intelligence from Open Sources (EIOS). World Health Organization (Current initiative page). https://www.who.int/initiatives/eios. Near-real-time detection and assessment of public-health threats from large open-source data volumes.

Provenance boundary

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